

Problem 1: Write and run a shell script to add two numbers given by user.

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem1.sh
#!/bin/bash

echo "Enter first number:"
read num1
echo "Enter second number:"
read num2

sum=$((num1 + num2))
echo "Sum = $sum"

silverlion@silverlion:~/Lab2$ ./problem1.sh
Enter first number:
6
Enter second number:
9
Sum = 15
silverlion@silverlion:~/Lab2$
```

Problem 2: Write a shell script to check whether a given number is even or odd

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem2.sh
#!/bin/bash

echo "Enter a number:"
read num

if [ $((num % 2)) -eq 0 ]
then
    echo "Even number"
else
    echo "Odd number"
fi

silverlion@silverlion:~/Lab2$ ./problem2.sh
bash: ./problem2.sh: No such file or directory
silverlion@silverlion:~/Lab2$ ./problem2.sh
Enter a number:
13
Odd number
silverlion@silverlion:~/Lab2$
```

Problem 3: Write a shell script to print the sum of all numbers from 1 to 100 using a for loop

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem3.sh
#!/bin/bash

sum=0

for ((i=1; i<=100; i++))
do
    sum=$((sum + i))
done

echo "Sum from 1 to 100 = $sum"
silverlion@silverlion:~/Lab2$ ./problem3.sh
Sum from 1 to 100 = 5050
silverlion@silverlion:~/Lab2$
```

Problem 4: Write a shell scrip to calculate the factorial of a number using a while loop

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem4.sh
#!/bin/bash

echo "Enter a number:"
read num

fact=1
i=1

while [ $i -le $num ]
do
    fact=$((fact * i))
    i=$((i + 1))
done

echo "Factorial = $fact"
silverlion@silverlion:~/Lab2$ ./problem4.sh
Enter a number:
5
Factorial = 120
silverlion@silverlion:~/Lab2$
```

Problem 5: Write a shell scrip that keeps taking user input until the word “exit” is entered

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem5.sh
#!/bin/bash

input=""

until [ "$input" = "exit" ]
do
    echo "Type something (type 'exit' to stop):"
    read input
done

echo "Program terminated."
silverlion@silverlion:~/Lab2$ ./problem5.sh
Type something (type 'exit' to stop):
Abrar_is_my_name exit
Type something (type 'exit' to stop):
exit
Program terminated.
silverlion@silverlion:~/Lab2$
```

Problem 6: Write a shell script to check if a file exists in the current directory

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem6.sh
#!/bin/bash

echo "Enter file name:"
read filename

if [ -f "$filename" ]
then
    echo "File exists."
else
    echo "File does not exist."
fi
silverlion@silverlion:~/Lab2$ ./problem6.sh
Enter file name:
problem4.sh
File exists.
silverlion@silverlion:~/Lab2$ ./problem6.sh
Enter file name:
problem10.sh
File does not exist.
silverlion@silverlion:~/Lab2$
```

Problem 7: Write a shell script to display the largest of three numbers entered by the user

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem7.sh
#!/bin/bash

echo "Enter first number:"
read a
echo "Enter second number:"
read b
echo "Enter third number:"
read c

if [ $a -ge $b ] && [ $a -ge $c ]
then
    echo "Largest = $a"
elif [ $b -ge $a ] && [ $b -ge $c ]
then
    echo "Largest = $b"
else
    echo "Largest = $c"
fi

silverlion@silverlion:~/Lab2$ ./problem7.sh
Enter first number:
19
Enter second number:
20
Enter third number:
21
Largest = 21
silverlion@silverlion:~/Lab2$
```

Problem 8: Write a shell script to print all even numbers between 1 and 50.

Pseudo Code:

```
silverlion@silverlion: ~/Lab2
silverlion@silverlion:~/Lab2$ cat problem8.sh
#!/bin/bash

for ((i=1; i<=50; i++))
do
    if [ $((i % 2)) -eq 0 ]
    then
        echo $i
    fi
done

silverlion@silverlion:~/Lab2$ ./problem8.sh
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
silverlion@silverlion:~/Lab2$
```